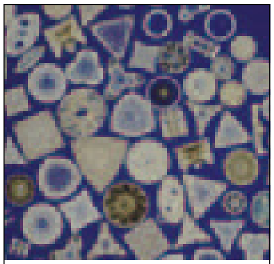
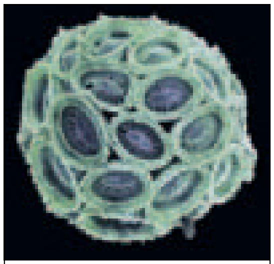
PHYTOPLANKTON



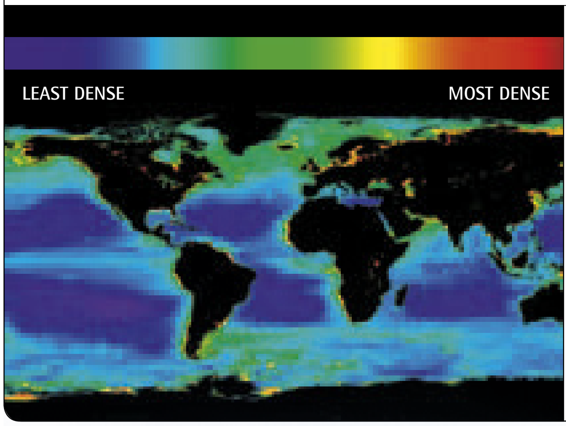
DIATOMS
In cool waters most of the phytoplankton consists of diatoms. These have shells of silica, which is basically glass. The shells fit together like microscopic boxes with lids, and they exist in a dazzling variety of forms. Most of the silicaceous ooze on the ocean floor consists of these silica shells.



COCCOLITHOPHORES
These tropical members of the phytoplankton have tough skeletons built up from tiny, ornamented disks of calcium carbonate or lime. When the organisms die and break up, the disks accumulate on the ocean floor as calcareous ooze. Over millions of years this may harden into limestone or chalk.

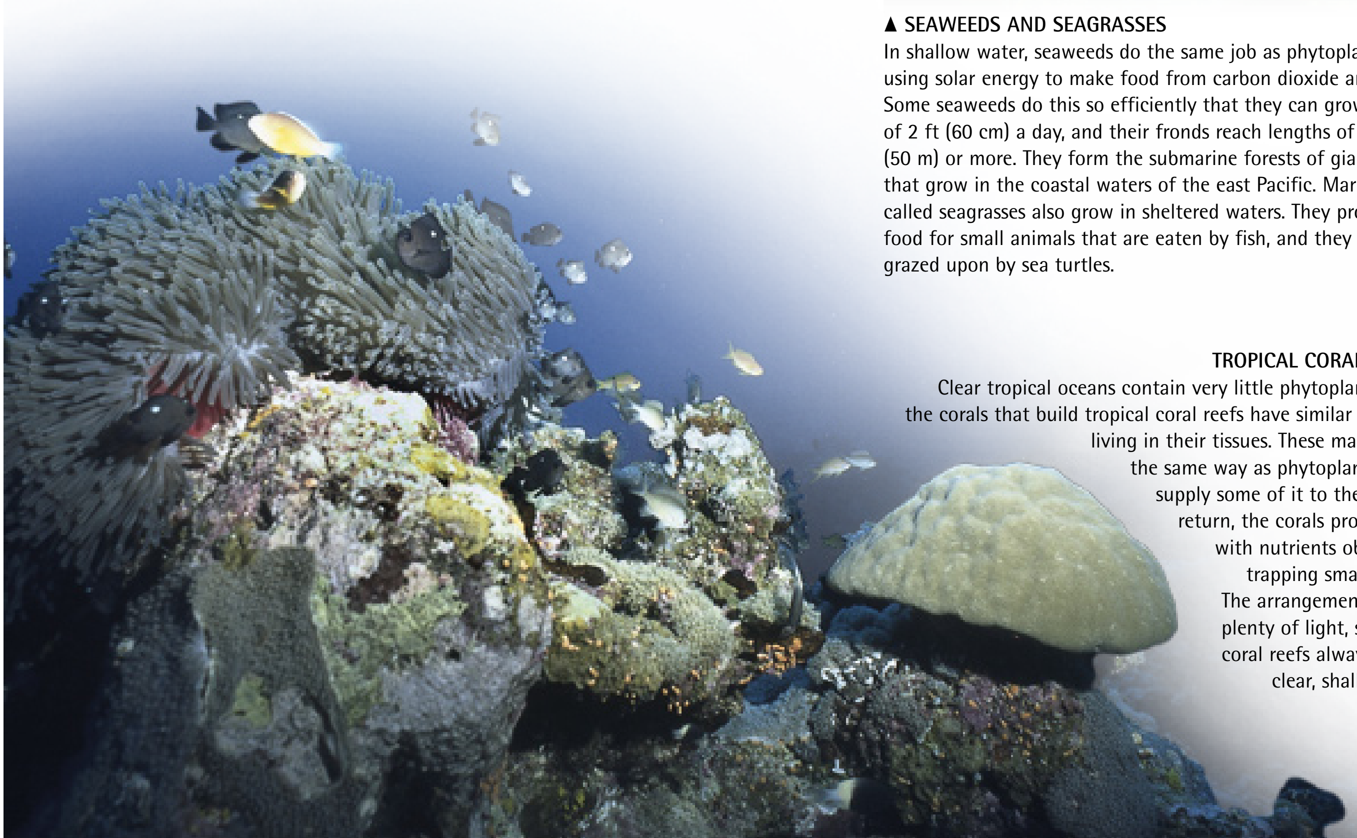
DINOFLAGELLATES
Like many microscopic organisms, dinoflagellates share features with both animals and plants. They are able to swim using tiny threadlike flagellae, but they make food by photosynthesis, like plants. Although they are a different shape, they have glassy silica skeletons, like diatoms.

PHYTOPLANKTON DENSITY



In tropical oceans, a layer of warm water stops nutrients reaching the sunlit surface zone, so phytoplankton cannot thrive. Upwelling currents overcome this near coasts, and in cooler oceans winter storms mix the water so phytoplankton can flourish. This satellite data shows that these waters are much richer than those of warm tropical oceans.

▲ SEAWEEDS AND SEAGRASSES
In shallow water, seaweeds do the same job as phytoplankton, using solar energy to make food from carbon dioxide and water. Some seaweeds do this so efficiently that they can grow at rates of 2 ft (60 cm) a day, and their fronds reach lengths of 165 ft (50 m) or more. They form the submarine forests of giant kelp that grow in the coastal waters of the east Pacific. Marine plants called seagrasses also grow in sheltered waters. They provide food for small animals that are eaten by fish, and they are also grazed upon by sea turtles.



TROPICAL CORAL REEFS ◄
Clear tropical oceans contain very little phytoplankton, but the corals that build tropical coral reefs have similar organisms living in their tissues. These make food in the same way as phytoplankton, and supply some of it to the corals. In return, the corals provide them with nutrients obtained by trapping small animals. The arrangement relies on plenty of light, so tropical coral reefs always grow in clear, shallow water.